

Exact Minimum Field Partition for One-Step Prediction in a Finite Five-Agent ABCW Model

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Abstract

Which distinctions in a present state must be retained to predict a specified future exactly? We study this question on a finite reachable set of a deterministic five-agent ABCW model in which binary agent actions and strategies coevolve with a weighted directed influence field. The dataset is generated from four initial-field families and all 4,096 action-strategy initial conditions, yielding 56,536 transitions and 2,562 distinct current fields. We retain the current action and seek the coarsest field-only partition that uniquely determines the complete one-step-ahead field on every observed transition. Nine natural feature families and all 511 of their nonempty combinations provide strong approximate predictors but no exact nontrivial field compression. We therefore construct an incompatibility graph whose vertices are current fields and whose edges join fields that produce different next fields under a shared observed action. A proper 692-coloring gives a constructive upper bound. An independent anchor-action decomposition, followed by exact solution of the 50 nontrivial induced subgraphs, gives the matching lower bound. Within an anchor group, the update rule shows analytically that clipping can hide only a $-1/0$ field-deviation distinction and that such a distinction is exposed precisely when it lies in a winning agent's row. For all 50 nontrivial anchor graphs, matching clique and coloring certificates give $\chi = \omega$, with chromatic numbers distributed as 39 groups of value 2, eight of value 3, two of value 4, and one of value 7. Hence the minimum number of field classes is exactly 692. The result is scoped to the specified finite reachable set and prediction task; the precise scope is stated in the Introduction.

Keywords: agent-based model; predictive partition; state aggregation; graph coloring; coarse-graining; exact prediction

Introduction

In complex dynamical systems, a more detailed description of the present does not necessarily provide a more useful description for predicting the future. A complete microscopic state may be sufficient to determine future evolution, but not every distinction encoded in that state need be relevant to a specified predictive task.

The central question of this study is therefore:

Which distinctions in the present must be retained in order to predict the future?

Equivalently:

How much of the present can be forgotten without losing the specified future?

This question is related to coarse-graining, state aggregation, lumpability, bisimulation, causal states, and finite-state-machine minimization. Across these settings, the difficult step is not merely to compress a state description, but to certify that no prediction-relevant distinction has been discarded. Exact minimum partitions are therefore useful as finite ground-truth cases: they separate unavoidable predictive distinctions from those retained only by a chosen representation. The frameworks differ in what future information must be preserved and in whether the reduced description must itself evolve recursively.

The present study does not propose a new general theory encompassing these frameworks. Instead, it considers a finite and explicitly computable agent-based dynamical system and determines the minimum number of distinctions required to preserve a specified one-step prediction exactly.

The system studied here is a five-agent ABCW model. Each agent has a binary action and a binary strategy, while the influence relations among agents are represented by a time-dependent weight matrix W . Actions are updated by reference to local influence relations, and game outcomes feed back into both strategy adaptation and field evolution.

The payoff rule has a Minority-Game-type competitive structure. The ABCW model is not intended to reproduce the standard Minority Game itself. Rather, the rule is used to introduce an environment in which all agents cannot simultaneously succeed in the same sense. The relationship of this construction to the Minority Game and the El Farol problem is discussed in Section 2.

ABCW is therefore a finite agent system in which agent behavior and the interaction field coevolve. We represent the field by its deviation from the fully symmetric reference field W^* :

$$\Delta W = W - W^*.$$

For the five-agent model, we use four initial fields and exhaust all action and strategy initial conditions for each field, giving a total of 4,096 initial conditions. Each trajectory is followed until the complete state recurs. The resulting finite reachable dataset contains 56,536 transitions, 2,562 distinct current fields ΔW , and 11,202 distinct $(a, \Delta W)$ states.

Within this finite system, an observation is treated not merely as the selection of a variable but as a mapping

$$P : \Omega \longrightarrow Y,$$

which specifies which distinctions among complete states are retained and which are discarded.

A key distinction is that closure of the observed dynamics and prediction of a specified target are different requirements. For an observation P , requiring

$$P(X_t) \longrightarrow P(X_{t+1})$$

to be single-valued is a self-closure condition. By contrast, for a specified prediction target Z , requiring

$$P(X_t) \longrightarrow Z$$

to be single-valued is a target-prediction condition. The primary objective of this study is the latter. Specifically, we retain the current action a_t and replace the current field ΔW_t with a coarser field-class label. We then ask whether this compressed description still determines the complete one-step-ahead field ΔW_{t+1} uniquely.

Because the object of compression is the field description, the action a_t is retained as an uncompressed conditioning variable. We do not simultaneously optimize a compression of the action space. This field-only restriction is part of the scope of the minimization problem solved in this paper.

Let

$$B : \Delta W \longrightarrow \mathcal{C}$$

be a field-only mapping, and define

$$P_B(X_t) = (a_t, B(\Delta W_t)).$$

We require

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

to be single-valued over all observed transitions. We call the ability of an observation to determine the specified target without ambiguity **predictive sufficiency**.

We first examine field compression based on human-designed network features, including norms, local structural descriptors, and out-strength. Some of these features achieve high predictive performance. Within the candidate family tested, however, no feature combination both merges genuinely distinct values of ΔW and predicts the complete one-step-ahead field exactly.

This failure does not imply that exact prediction requires all 2,562 observed fields to remain distinct. It shows only that the tested feature family does not identify an exact nontrivial compression. We therefore remove the restriction that the partition must be specified in advance through human-designed features and instead derive the necessary distinctions directly from the observed dynamics.

The finite dataset can be read as a partially specified input-output relation

$$(\Delta W, a) \longmapsto \Delta W',$$

where ΔW is the current field, a is the action condition, and $\Delta W'$ is the complete one-step-ahead field. If two current fields produce different next fields under a common action condition, they cannot belong to the same field class.

Accordingly, we construct an incompatibility graph G whose 2,562 vertices are the distinct observed current fields. Two fields are connected by an edge if, under at least one action condition observed for both fields, they produce different next fields. A field-only partition that preserves exact one-step prediction is then equivalent to a proper coloring of G .

The reduced field-class label is not required to serve recursively as the internal state at the next time step. The target being preserved is the complete next field ΔW_{t+1} , rather than the successor class

label. Consequently, the recursive successor-closure condition encountered in classical minimization of incompletely specified finite-state machines is not imposed here. For this depth-1 prediction problem, the minimum number of field classes is therefore given directly by

$$|\text{Im } B_{\min}| = \chi(G).$$

Compatibility need not itself be assumed to form an equivalence relation. Pairwise distinctions that cannot be merged are encoded directly as incompatibility edges, and a proper coloring separates every incompatible pair. The relationship between this formulation and classical state-minimization problems is developed in Section 2.

The main result is

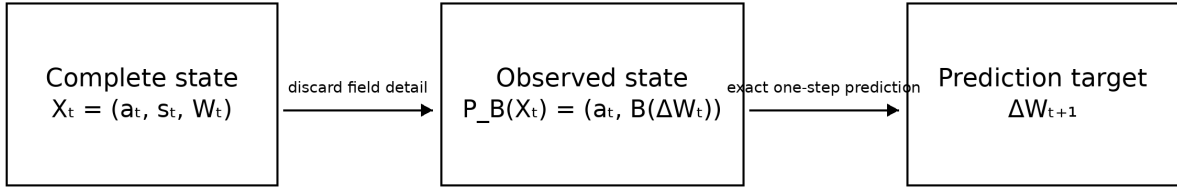
$$\chi(G) = 692.$$

Thus, for the finite reachable set analyzed here and for the specified task of exact one-step field prediction, the 2,562 distinct current fields can be compressed to 692 field classes:

$$2562 \longrightarrow 692,$$

while preserving exact prediction of the complete next field.

Figure 1 summarizes the predictive-partition problem and the resulting reduction.



Field partition: 2,562 observed fields $\rightarrow$ minimum 692 classes

Which distinctions in the present field are necessary for the specified future?

Figure 1: Predictive-partition problem. The complete ABCW state is compressed by retaining the action and replacing the full field deviation with a field-only partition label. The prediction target is the exact one-step-ahead field deviation. The main result reduces 2,562 observed fields to a minimum of 692 classes.

Measured by the number of field classes, the corresponding compression rate is

$$1 - \frac{692}{2562} \simeq 72.99\%.$$

This is a reduction in the number of field classes; it should not be interpreted as a 72.99% reduction in information content.

The value 692 is not merely the output of a coloring heuristic. A lower bound is obtained by fixing the anchor action

$$a = (-, -, -, -, -).$$

Under this action, the 1,239 observed current fields produce 625 distinct next fields, so at least 625 field classes are required. Within 50 of these 625 anchor groups, additional incompatibilities appear when other action conditions are taken into account. The chromatic numbers of the corresponding induced subgraphs are solved exactly by combining clique lower bounds with colorability searches. These refinements require 67 additional classes, giving the lower bound

$$625 + 67 = 692.$$

For the matching upper bound, we explicitly construct a proper 692-coloring of the incompatibility graph. Mapping this coloring back to all 56,536 observed transitions verifies that every observed pair $(a_t, B(\Delta W_t))$ determines a unique ΔW_{t+1} . The upper and lower bounds therefore coincide, establishing $\chi(G) = 692$ exactly.

Scope statement. The number 692 is the exact minimum for the finite five-agent reachable set studied here, with the current action retained, only the field compressed, and the complete one-step-ahead field used as the target. It is not asserted as a universal ABCW macrostate count; formal limitations are collected in Section 8.

The broader point is that microscopic distinctness and predictive relevance need not coincide. In this finite ABCW system, fields that are microscopically different can sometimes be assigned to the same predictive class without losing the specified future. At the same time, simple network features and geometric proximity between fields do not fully identify which distinctions must be preserved. Once the dynamics, reachable set, retained conditioning variables, and prediction target are specified, the required distinctions can instead be posed directly as a minimum-partition problem.

The resulting partition occupies an intermediate descriptive level between a complete microscopic representation and a small set of simple macroscopic statistics: it retains exactly the distinctions required by the specified prediction task on the observed finite system. This does not by itself make the 692 classes an interpretable set of macroscopic variables, but it provides an exact target against which candidate coarse descriptions can be evaluated.

The contributions of this study are threefold.

First, for a finite ABCW system, we distinguish self-closure from target prediction and formulate the question of which present distinctions must be retained as an explicit observation-minimization problem.

Second, we distinguish compression by preselected network features from a predictive partition derived from future responses, and we formulate the resulting field-only, action-conditioned, depth-1 problem as a graph-coloring problem.

Third, for the finite reachable set considered here, we establish that the exact minimum number of field classes is 692 by providing both a constructive upper bound and an independent matching lower bound. We also derive analytically the within-anchor mechanism that generates incompatibility: the anchor update can hide only a $-1/0$ field-deviation distinction, and another co-observed action reveals that distinction precisely through a winning agent’s row. For all 50 nontrivial anchor graphs, matching clique and coloring certificates give $\chi = \omega$, explaining the additional 67 classes as explicitly forced predictive distinctions rather than an artifact of a coloring heuristic.

These contributions do not assert novelty for the general theories of state minimization, lumpability, bisimulation, or causal states. Rather, the study formulates a specifically scoped prediction problem for a finite coevolving agent–field system and solves its exact minimum on a fully inspectable dataset.

The remainder of the paper is organized as follows. Section 2 positions the study in relation to the El Farol problem, the Minority Game, state aggregation, lumpability, bisimulation, computational mechanics, and minimization of incompletely specified finite-state machines. Section 3 defines the ABCW model and the finite reachable dataset. Section 4 formalizes observation, self-closure, predictive sufficiency, and the prediction target used in this paper. Section 5 evaluates compression based on natural network features. Section 6 constructs the incompatibility graph and determines the exact minimum field-only partition. Section 7 analyzes the internal structure of the 692 classes. Section 8 discusses the interpretation, scope, and limitations of the result. Section 9 concludes the paper.

Related Work and Positioning

Competitive agent models: El Farol and the Minority Game

The competitive payoff structure of the ABCW model shares part of its motivation with the El Farol Bar problem and the Minority Game.

The El Farol Bar problem introduced by Arthur (1994) considers a setting in which several agents seek access to the same limited opportunity or resource, so that the desirability of an action depends on the choices of other agents. Attendance is advantageous when relatively few agents attend, but its value declines when too many make the same choice. Consequently, the rationality of an action cannot be fixed independently of the behavior of the population.

The Minority Game introduced by Challet and Zhang (1997) formulates this competitive structure as a simplified binary-choice game. At each time step, agents select one of two alternatives, and agents in the minority receive a payoff. The value of an action is therefore not intrinsic to that action but depends on how many other agents choose it.

ABCW employs this Minority-Game-type competition. Each of five agents takes a binary action, with agents in the minority designated as winners and those in the majority as losers. This creates a competitive payoff environment in which all agents cannot simultaneously succeed in the same sense.

ABCW is not intended, however, to reproduce or extend the standard Minority Game, nor is it designed to analyze its characteristic collective phenomena. Standard Minority Game research

commonly focuses on agents' strategy sets, information histories, adaptation, collective efficiency, and aggregate fluctuations. By contrast, the central object in the present study is the time-dependent field W representing influence relations among agents.

In ABCW, game outcomes feed back not only into agent scores but also into the evolution of the field. The Minority-Game-type rule is therefore a component that generates the coupled process

$$\text{action} \longrightarrow \text{outcome} \longrightarrow \text{field update} \longrightarrow \text{next action},$$

rather than the final object of analysis. The primary question is not which strategy dominates in a Minority Game or how efficiently the population coordinates. Instead, for the field dynamics generated by this competitive agent system, we ask:

Which distinctions among current fields must be retained to predict the complete one-step-ahead field exactly?

The El Farol problem and the Minority Game thus provide the agent-modeling background for the competitive dynamics of ABCW. They are not the direct theoretical basis of the predictive state-reduction problem addressed here. The more immediate connections are to state aggregation and finite-state minimization.

State aggregation and the question of relevant distinctions

The central question of this study—which distinctions in the present must be retained for prediction—belongs to a broad class of problems concerned with aggregating a state space while preserving properties required for dynamics or prediction.

A classical example is lumpability in Markov chains. In the formulation of Kemeny and Snell (1960), a partition of the state space into lumps yields a closed Markov process on the aggregated states when, for any two states in the same lump, the transition probabilities into every destination lump coincide. Lumpability therefore provides conditions under which microscopic states can be merged while retaining well-defined reduced dynamics.

The self-closure condition introduced in Section 4 has a related motivation in deterministic finite dynamics. If

$$P(X) = P(X')$$

implies

$$P(\Phi(X)) = P(\Phi(X')),$$

then two complete states assigned to the same observed state also evolve to the same observed state at the next time step. Here X is a complete state, Φ is the one-step update map on complete states, and $P : \Omega \rightarrow Y$ is a general observation map on the complete state space. This general map P should be distinguished from the field-only map $B : \Delta W \rightarrow \mathcal{C}$ that is later minimized.

Self-closure is not, however, the primary requirement in this paper. We distinguish prediction of the next observed state,

$$P(X_{t+1}),$$

from prediction of a separately specified target,

$$Z = g(X_{t+1}).$$

The required condition is therefore not necessarily closure of

$$P(X_t) \longrightarrow P(X_{t+1}),$$

but single-valuedness of

$$P(X_t) \longrightarrow Z.$$

This distinction between closed reduced dynamics and sufficient information for a specified target is the starting point of our state-aggregation problem.

Bisimulation and model minimization

The idea of merging states while preserving future behavior is also closely related to bisimulation and bisimulation-based model minimization.

Givan, Dean, and Greig (2003), for example, study equivalence notions for Markov decision processes and use bisimulation-based aggregation to construct reduced MDPs that preserve optimal policies of the original process. The common principle is that states are grouped not because they are superficially similar, but because their distinction is unnecessary for the future behavior that must be preserved.

The system and preservation target considered here are different. ABCW is not an MDP; on the stored finite reachable set, it has deterministic update rules. Nor do we minimize the complete ABCW state. Writing the complete state as

$$X = (a, s, W),$$

we retain the action a in the observation and compress only the field deviation ΔW through

$$B : \Delta W \longrightarrow \mathcal{C}.$$

Moreover, the preserved target is neither an optimal policy nor a reward structure nor the full reduced dynamics. It is the complete one-step-ahead field ΔW_{t+1} . We therefore do not describe the resulting minimum partition as a bisimulation quotient.

Computational mechanics and causal states

Computational mechanics provides another closely related predictive perspective. Shalizi and Crutchfield (2001) define causal states by identifying past histories that induce the same conditional probability distribution over futures. The resulting ϵ -machine is characterized as a minimal representation retaining the information required for prediction.

The conceptual connection to the present study is clear: distinctions among complete histories or microscopic states are retained only when they matter for the future. Similarly, the 692 classes derived here are defined not by static similarity in network structure but by the one-step field responses generated under action conditions.

The two constructions should nevertheless not be identified. Causal states are, in principle, equivalence classes of histories defined by conditional distributions over future sequences. The present study instead partitions current fields ΔW in a finite reachable set, with the prediction target restricted to the complete one-step-ahead field ΔW_{t+1} .

The ABCW data are also partially specified: not every action condition is observed for every current field. As a result, the relation “no conflict occurs under the observed common conditions” need not be a transitive equivalence relation. We therefore do not call the 692 classes causal states or an ϵ -machine. The relevant connection is the shared objective of preserving future-relevant distinctions.

Incompletely specified finite-state machine minimization

The most direct mathematical correspondence is with minimization of incompletely specified finite-state machines (Paull and Unger 1959; Kohavi and Jha 2009). When outputs or transitions are missing for some state-input pairs, reduction is formulated through compatibility rather than ordinary equivalence. Classical recursive machines additionally require successor classes to satisfy closure conditions.

The ABCW records can likewise be viewed as a partial input-output table

$$(\Delta W, a) \mapsto \Delta W'.$$

If two fields are observed under the same action but produce different next fields, they cannot share a predictive class. Encoding every such pair as an incompatibility edge makes an exact field partition a proper coloring, and hence

$$|\text{Im } B_{\min}| = \chi(G).$$

The key difference from classical FSM minimization is that the reduced field label is not required to act recursively as the next internal state: the preserved target is the complete one-step-ahead field. Thus successor-class closure is unnecessary for this depth-1 problem, and nontransitive compatibility in the partially specified data can be handled directly by pairwise graph constraints.

Feature-based coarse-graining and predictive partitioning

We further distinguish two approaches to state compression.

The first computes a set of preselected features from the state:

$$\Delta W \longrightarrow B_{\text{feature}}(\Delta W).$$

Examples considered here include norms, local structural descriptors, hub structure, and out-strength. The advantage of this approach is interpretability: the resulting observables have meanings chosen in advance. A low-dimensional feature representation, however, does not necessarily constitute genuine coarse-graining. On a finite state set, even a low-dimensional feature vector may uniquely identify every microscopic state. When features do merge states, they may also discard distinctions required for prediction.

The second approach does not choose the features in advance. Instead, it derives the required distinctions from future responses, conceptually through the response structure

$$\Delta W \longrightarrow (a \longmapsto \Delta W') .$$

The 692-class partition belongs to this second approach. Its classes are not guaranteed to admit an immediate interpretation in terms of simple network statistics. In return, the partition is determined directly by the requirement that the specified predictive target be preserved.

We refer to this property as **predictive sufficiency**. An observation P is predictively sufficient for a target Z on the dataset under consideration if

$$P(X_t) \longrightarrow Z$$

is single-valued, so that Z is determined without ambiguity by $P(X_t)$. Predictive sufficiency does not mean merely that a state can be expressed with a small number of variables. The question is instead how many distinctions can be discarded while retaining all distinctions required for the specified target.

Predictive state representations. Predictive state representations (PSRs) represent dynamical state through a collection of action-conditional predictions of future observations (Littman, Sutton, and Singh 2001). This is conceptually close to the present emphasis on prediction-relevant distinctions: in both cases the usefulness of a state description is tied to what it determines about future observations. The constructions are nevertheless different. PSRs are recursive predictive representations, typically built from multi-step tests, whereas the present problem fixes a finite reachable set, retains the current action as a conditioning variable, and asks for the minimum field-only partition preserving a specified complete one-step-ahead field. We therefore treat PSRs as a neighboring predictive-state perspective rather than as the minimization method used here.

Position of the present study

ABCW uses a Minority-Game-type competitive rule to generate coupled agent-field dynamics, but the paper does not analyze standard Minority Game phenomena as an end in itself. Its state-reduction problem is also narrower than the neighboring general frameworks reviewed above: the current action is retained, the field alone is compressed, and the preserved object is a specified one-step output on a finite partially specified reachable set.

The contribution is therefore concrete rather than foundational: we formulate this depth-1 predictive minimization as an incompatibility graph, establish $\chi(G) = 692$ by independent matching upper and lower bounds, and derive the clipping/winner-row mechanism responsible for the nontrivial within-anchor conflicts. The example is fully inspectable and gives an exact finite case in which microscopic state-distinctness can be separated from distinctions required for a specified prediction.

ABCW Model and Dataset

This section defines the five-agent ABCW model analyzed in this paper and the finite reachable dataset used in the predictive-partition problem. The purpose is not to reconstruct the exploratory development of ABCW, but to fix the state variables, update rules, field representation, and scope of the subsequent analysis.

State Variables and Interaction Network

Let the set of agents be

$$V = \{1, \dots, n\},$$

with

$$n = 5$$

throughout this paper. At time t , each agent i has a binary action

$$a_i(t) \in \{-1, +1\}$$

and a binary strategy

$$s_i(t) \in \{-1, +1\}.$$

The value $s_i(t) = +1$ denotes a trend-following strategy that follows the reference signal, whereas $s_i(t) = -1$ denotes a contrarian strategy that acts against it.

Influence relations among agents are represented by a fixed topology

$$E = (E_{ij}), \quad E_{ij} \in \{0, 1\}$$

and a time-dependent nonnegative weight matrix

$$W(t) = (w_{ij}(t)).$$

The condition $E_{ij} = 1$ indicates that the directed edge $i \rightarrow j$ is permitted by the topology, and $w_{ij}(t)$ represents the strength of the influence exerted by agent i on agent j at time t . If its weight reaches zero, the topological edge itself is not removed as long as $E_{ij} = 1$.

Because strategy is included as a state variable, the complete state is defined as

$$X(t) = (a(t), s(t), W(t)).$$

Table 1 summarizes the principal variables and their roles.

Table 1. ABCW variables and update-rule summary

Symbol	Meaning	Domain / type	Role / update
$a_i(t)$	Action	$\{-1, +1\}$	Current binary action
$s_i(t)$	Strategy	$\{-1, +1\}$	Trend-following / contrarian; loser flips strategy
$u_i(t)$	Payoff	$\{-1, +1\}$ for $n = 5$	Minority +1, majority -1; all-same gives -1
$W(t)$	Influence field	Nonnegative weighted directed matrix	Outgoing edges updated by source payoff
$\sigma_i(t)$	Reference signal	$[-1, 1]$	Mean action among strongest effective incoming references
$\Delta W(t)$	Field deviation	Matrix	$W(t) - W^*$

Payoff, Reference Signal, and Update Rules

Minority-game payoff

At each time step, every agent chooses either -1 or $+1$. Agents in the minority are designated winners and those in the majority losers. The payoff is

$$u_i(t) \in \{-1, +1\},$$

with $u_i(t) = +1$ for a minority agent and $u_i(t) = -1$ for a majority agent. Because $n = 5$ is odd, a tie between the two binary actions is impossible. If all agents choose the same action, all are treated as losers and receive $u_i(t) = -1$.

This payoff rule is not intended to reproduce the full standard Minority Game. It is used to introduce into ABCW a competitive condition under which all agents cannot win simultaneously.

Reference signal

The set of agents that agent i can reference is

$$\mathcal{N}_i = \{j \in V \mid E_{ji} = 1\}.$$

Define the maximum incoming weight available to agent i as

$$m_i(t) = \begin{cases} \max_{j \in \mathcal{N}_i} w_{ji}(t), & \mathcal{N}_i \neq \emptyset, \\ 0, & \mathcal{N}_i = \emptyset \end{cases}.$$

When $m_i(t) > 0$, let

$$M_i(t) = \{j \in \mathcal{N}_i \mid w_{ji}(t) = m_i(t)\}$$

be the set of agents attaining that maximum, and define the reference signal as

$$\sigma_i(t) = \frac{1}{|M_i(t)|} \sum_{j \in M_i(t)} a_j(t).$$

If several agents share the maximum incoming weight, no single agent is selected arbitrarily; all tied agents are referenced with equal weight. If their actions balance, then $\sigma_i(t) = 0$. We also set $\sigma_i(t) = 0$ when no referenceable agent exists or when the maximum available influence is zero, treating both cases as the absence of an effective reference signal.

The game outcome $u_i(t)$ is thus determined by the actions of all agents, whereas the reference information $\sigma_i(t)$ used for the next action is determined by local influence relations.

Strategy adaptation

In the adaptive ABCW model used here, each agent updates its strategy using only the outcome of the immediately preceding game. A winner retains its strategy, whereas a loser reverses it. Thus,

$$s_i(t+1) = \begin{cases} s_i(t), & u_i(t) = +1, \\ -s_i(t), & u_i(t) = -1 \end{cases}$$

or, equivalently,

$$s_i(t+1) = u_i(t)s_i(t).$$

Action update

After payoffs have been calculated from the current actions and reference signals from the current field, strategies are updated first. A losing agent uses this **updated strategy**, rather than its previous strategy, to determine its next action. The action update is therefore

$$a_i(t+1) = \begin{cases} a_i(t), & u_i(t) = +1, \\ a_i(t), & u_i(t) = -1, \sigma_i(t) = 0, \\ \text{sgn}(s_i(t+1)\sigma_i(t)), & u_i(t) = -1, \sigma_i(t) \neq 0 \end{cases}.$$

A winner retains its current action. A loser also retains its current action if no effective reference signal is available; otherwise, it selects the next action according to its updated strategy and the reference signal.

Field update

Finally, the payoff is fed back into the field. Let the learning rate be $\eta > 0$. All experiments analyzed in this paper use

$$\eta = 1.$$

The field update is

$$w_{ij}(t+1) = \begin{cases} \max(0, w_{ij}(t) + \eta u_i(t)), & E_{ij} = 1, \\ 0, & E_{ij} = 0 \end{cases}.$$

Thus, when agent i wins, the weights of the existing outgoing edges from that agent increase; when it loses, they decrease, subject to the lower bound of zero.

The update order for one game is fixed as

$$\begin{array}{c}
(a(t), s(t), W(t)) \\
\downarrow \\
u(t), \sigma(t) \\
\downarrow \\
s(t+1) \quad . \\
\downarrow \\
a(t+1) \\
\downarrow \\
W(t+1)
\end{array}$$

This ordering is part of the model definition because losing agents use their updated, rather than previous, strategies when selecting their next actions.

Every branch of the payoff, reference, strategy, action, and field updates is resolved by these rules. In particular, tied reference agents are averaged rather than selected arbitrarily, a zero reference signal invokes the action-retention rule above, and no random tie-breaking or stochastic choice is used. Consequently, for fixed $(a(t), s(t), W(t))$ and E , the complete successor state is unique. The ABCW system analyzed here is therefore deterministic, and repeated runs from the same initial state generate the same finite trajectory and transition records.

The dataset uses the deterministic update order specified above: payoff and reference signal, strategy, action, then field.

Field Representation

For the five-agent system, define the fully symmetric reference field as

$$W^* = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

The reference field W^* is not assumed to be an equilibrium or a naturally privileged social state. It serves only as a reference point for describing field deformation.

The field deviation is defined as

$$\Delta W(t) = W(t) - W^*,$$

or componentwise as

$$\Delta w_{ij}(t) = w_{ij}(t) - w_{ij}^*.$$

Thus, $\Delta W(t) = 0$ means that the current field coincides with the reference field. A nonzero deviation $\Delta W \neq 0$ does not necessarily imply matrix asymmetry $W \neq W^T$. Throughout this paper, *field*

deviation refers to the difference from the reference field itself. This ΔW is the field variable used in the subsequent prediction problem.

Dataset Construction

The analysis uses four initial fields for the five-agent ABCW model:

- Baseline
- Hub
- Local
- Hub+Local

For Baseline, $W(0) = W^*$. In each of the other three initial fields, four directed edges whose reference weight is 1 are strengthened to 4, while the update rules remain unchanged. Only the placement of these strong edges differs. Numbering the agents $1, \dots, 5$ and denoting the directed edge $i \rightarrow j$ by (i, j) , the strengthened edges are:

- Hub: $(1, 2), (1, 3), (1, 4), (1, 5)$
- Local: $(1, 2), (2, 3), (3, 4), (4, 5)$
- Hub+Local: $(1, 3), (1, 5), (2, 3), (4, 5)$

In Hub, the four outgoing edges from agent 1 to the other agents are strengthened. Local strengthens a chain of four edges. Hub+Local combines two outgoing edges from agent 1 with two local competing edges that share their destination agents.

Each of these three non-baseline fields therefore has four nonzero initial deviation components and the same Frobenius norm,

$$\|\Delta W(0)\|_F = \sqrt{4 \times 3^2} = 6.$$

These configurations are not intended as an exhaustive classification of network structures. They serve as initial-condition families from which a finite reachable state set is constructed.

For each initial field, we exhaust all combinations of the five-agent initial action vector $a(0)$ and initial strategy vector $s(0)$. Because every component is binary, this gives

$$2^5 \times 2^5 = 1024$$

initial conditions per field and

$$4 \times 1024 = 4096$$

initial conditions in total.

Each case is evolved until the complete state recurs, and every transition encountered before recurrence is collected. This procedure yields 56,536 transitions containing 2,562 distinct current fields ΔW and 11,202 distinct $(a, \Delta W)$ states.

Table 2 summarizes the dataset construction and the resulting counts.

Table 2. Dataset construction and summary

Quantity	Value
Agents	5
Learning rate η	1
Initial field configurations	4
Action configurations per field	32
Strategy configurations per field	32
Initial conditions	4096
Observed transitions	56536
Distinct current fields	2562
Distinct $(a, \Delta W)$ states	11202

The 56,536 transitions do not represent the complete theoretical state space of ABCW. All subsequent results should be interpreted as statements about the stored finite transition set reached from the four initial fields and 4,096 initial conditions defined above. We make no claim of generalization to unreachable states, other numbers of agents, or other initial fields. Nor do we interpret the recurrence of every trajectory among the 4,096 cases as a proof that the weights are bounded in general.

Observation and Predictive Objective

This section formulates, for the finite ABCW system defined above, the problem of determining which distinctions in the present must be retained to predict a specified future exactly. The central point is to distinguish closure of the observed dynamics from sufficiency for a particular prediction target.

Observation as Information Loss

Let Ω denote the complete state space, and represent an observation as a mapping

$$P : \Omega \longrightarrow Y.$$

If two distinct complete states $X, X' \in \Omega$ satisfy

$$P(X) = P(X'),$$

then the observation P discards the distinction between them and treats them as the same observed state. An observation is therefore not merely a choice of which variables to inspect. It is a mapping that determines which distinctions among complete states are retained and which are identified.

The complete ABCW state is

$$X = (a, s, W).$$

Because the reference field W^* is fixed, W and ΔW are in one-to-one correspondence. For example,

$$P_W(X) = \Delta W$$

discards distinctions in action a and strategy s , whereas

$$P_{aW}(X) = (a, \Delta W)$$

discards only distinctions in strategy s .

Self-Closure and Target Prediction

Write the deterministic ABCW update as

$$X_{t+1} = \Phi(X_t).$$

An observation P is called **self-closed** if

$$P(X) = P(X')$$

always implies

$$P(\Phi(X)) = P(\Phi(X')).$$

Under this condition, the next observed state is uniquely determined by the current observed state. Self-closure is a deterministic quotient-consistency condition that makes the coarse-grained transition well defined. It is related to aggregation conditions in lumpability and bisimulation, but it is not identical to causal-state construction from conditional distributions over future sequences (Shalizi and Crutchfield 2001; Givan, Dean, and Greig 2003).

Self-closure is distinct from sufficiency for a specified prediction target. If the quantity of interest is

$$Z = g(X_{t+1}),$$

the required condition is that

$$P(X_t) \longrightarrow Z$$

be single-valued. It is not necessary for

$$P(X_t) \longrightarrow P(X_{t+1})$$

to be single-valued.

This distinction appears directly in the finite ABCW dataset. For the field-only observation

$$P_W(X) = \Delta W,$$

the next field is unique for 1,390 of the 2,562 observed current fields:

$$\frac{1390}{2562} \simeq 0.54254.$$

Thus, the current field alone determines the complete next field for approximately 54.3% of the observed current fields, but not in general.

When the current action is retained through

$$P_{aW}(X) = (a, \Delta W),$$

the relation

$$(a_t, \Delta W_t) \longrightarrow \Delta W_{t+1}$$

is single-valued for all 11,202 observed $(a, \Delta W)$ states. By contrast, the next state of this observation itself,

$$(a_t, \Delta W_t) \longrightarrow (a_{t+1}, \Delta W_{t+1}),$$

is single-valued for only 8,390 of the 11,202 observed states:

$$\frac{8390}{11202} \simeq 0.748973.$$

This is approximately 74.9%. An observation that is not self-closed can therefore still predict a specified component of the future perfectly. We keep these two requirements separate and fix the prediction target explicitly below.

One-Step Exact-Field Prediction

The prediction target in this paper is

$$Z = \Delta W_{t+1}.$$

That is, the observation must predict the **complete one-step-ahead field deviation** without error.

Retaining $(a_t, \Delta W_t)$ makes this target single-valued throughout the finite dataset, but it also retains all 2,562 distinct field deviations. We therefore keep the current action a_t unchanged while further coarsening only the field component.

Let

$$B : \Delta W \longrightarrow \mathcal{C}$$

be a field-only map, and define the corresponding observation as

$$P_B(X_t) = (a_t, B(\Delta W_t)) .$$

The map B satisfies the predictive requirement if, for any two current states X_t and X'_t in the analyzed transitions,

$$a_t = a'_t$$

and

$$B(\Delta W_t) = B(\Delta W'_t)$$

imply

$$\Delta W_{t+1} = \Delta W'_{t+1} .$$

Equivalently, we require

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

to be single-valued on the stored finite transition set.

The action a is not itself compressed, and the prediction target is not coarse-grained: the required output remains the complete next field deviation. The problem considered here is therefore an **action-conditioned, one-step, exact-output, field-only predictive-partition problem**.

We call the required single-valuedness **predictive sufficiency** for the specified target. In the present formulation, B is admissible when $(a_t, B(\Delta W_t))$ is predictively sufficient for ΔW_{t+1} on the analyzed dataset.

Minimum Field Partition

Field deviations mapped to the same value by B are treated as members of the same class, and the set of classes is $\text{Im } B$. The complete field observation has

$$|\text{Im } B| = 2562.$$

If exact predictive accuracy can be retained with

$$|\text{Im } B| < 2562,$$

then distinct field deviations can be identified for the specified prediction task, at least on the finite reachable set considered here.

We define the optimization problem as

$$\min_B |\text{Im } B|$$

subject to

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1} \quad \text{being single-valued on all observed transitions.}$$

This formulation is related in motivation to the Information Bottleneck, which compresses a representation while retaining relevant information. It is not the standard Information Bottleneck objective. Rather than optimizing a trade-off between compression and relevant information through mutual information, we impose zero prediction error as a hard constraint and minimize the finite-set class count $|\text{Im } B|$ (Tishby, Pereira, and Bialek 2000).

The problem also differs from finding the coarsest self-closed partition. Here, the retained action serves as a conditioning variable, and the quantity preserved is the external target ΔW_{t+1} . The minimization is therefore defined by target prediction rather than self-closure of the observation.

When useful, we report the field-class compression rate

$$C_{\Delta W} = 1 - \frac{|\text{Im } B|}{2562}.$$

This is a class-count-based compression rate. It is not a reduction rate in Shannon information or in the number of bits required for representation.

This formulation separates the minimization problem from the prior selection of features that appear intuitively natural. Section 5 first evaluates baselines constructed from natural network features. Section 6 then derives the required field distinctions directly from the predictive constraint and determines the minimum partition.

Natural-Feature Baselines

Before deriving the minimum predictive partition directly from the dynamics, we examine how coarsely the current field ΔW can be represented by human-selected network features, including norms, local structure, hub structure, and out-strength. The purpose is to assess whether interpretability, genuine field compression, and predictive sufficiency for the complete one-step-ahead field can be achieved simultaneously.

Natural-feature compression and its limit

The out-strength vector, for example, achieves very high predictive performance. Its state-level determinism score is evaluated over the observed states defined by $(a, \text{out-strength})$. The next field is unique for 11,084 of the 11,142 distinct observed states:

$$\frac{11084}{11142} = 0.994794 \dots$$

The resulting state-level determinism score is 99.4794%. The remaining 58 observed states are conflicts: more than one next field occurs from the same observed state.

The denominator here is not the total number of 56,536 transitions but the 11,142 distinct observed states. This state-level score should therefore be distinguished from evaluations weighted over the complete transition set.

We then searched for simple feature combinations that would resolve these 58 conflicts while genuinely compressing ΔW . Within the candidate family examined, no feature representation simultaneously achieved nontrivial field compression and exact predictive determinism.

This result contrasts with the response-based minimum predictive partition constructed exactly in [Section 6](#). That partition compresses 2,562 fields into 692 classes while preserving exact one-step field prediction across all 56,536 observed transitions.

The contrast reflects two different approaches to compression. The first summarizes the field using features selected in advance:

$$\Delta W \longrightarrow B(\Delta W).$$

The second derives the necessary distinctions from equivalence of future responses:

$$\Delta W \longrightarrow (a \longmapsto \Delta W').$$

The first approach favors interpretability but may discard information required for prediction. The second preserves distinctions required by the predictive task, but the resulting classes need not have a simple interpretation as network statistics. The 692-class partition derived in [Section 6](#) belongs to the second category.

Exhaustive combinations of the candidate feature families

The exhaustive search uses nine feature maps of the field matrix $D = \Delta W$: squared Frobenius norm, matrix rank, stable rank, the ordered singular-value vector, permutation-invariant out- and in-strength distributions, the lexicographically sorted joint in/out-strength distribution, and the ordered out- and in-strength vectors that retain agent identities. Writing $r(D) = D\mathbf{1}$, $c(D) = D^\top\mathbf{1}$, and $\sigma_1(D) \geq \dots \geq \sigma_5(D)$, these are standard quantities derived from $\|D\|_F$, rank, singular values, $r(D)$, and $c(D)$. Singular values and stable rank are canonicalized to ten significant digits; all other feature values are integer-valued. A feature combination is the tuple of its selected map values, with no learned weighting or post-processing.

These nine feature maps have

$$2^9 - 1 = 511$$

nonempty subsets. We exhaustively evaluated all 511 feature combinations. None simultaneously merged genuinely distinct values of ΔW and determined the complete one-step-ahead field without ambiguity.

The out-strength vector yields high determinism but leaves 58 conflicting observed states. For contrast, we also report B^* , the combination of squared Frobenius norm and ordered out-strength vector. It achieves exact prediction within the candidate family, but it uniquely distinguishes all 2,562 fields and therefore provides no genuine field compression; its role in Table 3 is to show that exactness alone does not imply compression.

Table 3 compares representative natural-feature baselines with the exact minimum partition. Here D_{state} is the state-level determinism score, whereas A_{freq} is the corresponding transition-frequency-weighted score.

Table 3. Natural-feature baselines and exact minimum partition

Observation	Field classes	Observed (a, B) states	D_{state}	A_{freq}	True field compression
Full field ΔW	2562	11202	1	1	No
Out-strength vector	2516	11142	0.994794	0.994658	Yes
Best true- compression candidate by D_{state} (in-strength distribution + out-strength vector)	2561	11200	0.999821	0.999717	Yes
B^* (Frobenius norm + out-strength vector)	2562	11202	1	1	No
Exact minimum partition B_{692}	692	8084	1	1	Yes

Figure 2 places all 511 candidates in the same performance–compression plane and highlights the exact minimum partition.

The failure of the natural-feature search to achieve both exact prediction and genuine compression does not imply that nearly all 2,562 current fields must remain distinct. It shows only that the tested feature family does not express the required partition. In the next section, we remove the restriction of selecting features in advance and derive the necessary distinctions directly from conflicts in future responses.

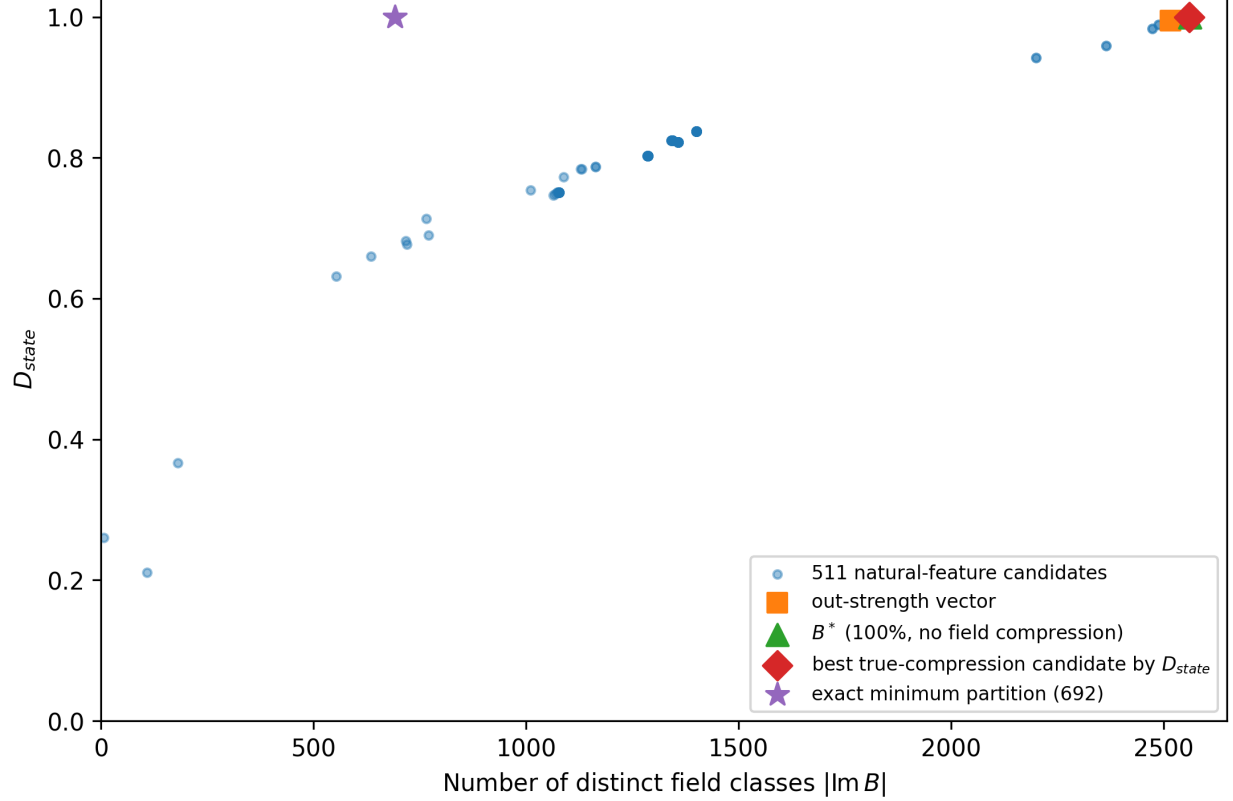


Figure 2: Prediction performance versus actual field compression. All 511 nonempty combinations from the nine natural feature families are plotted using the number of distinct field values and the state-level determinism score. The out-strength vector, the 100%-predictive but noncompressive B^* representation, the best genuinely compressive candidate by D_{state} , and the exact 692-class partition are highlighted.

Exact Minimum Predictive Partition

Incompatibility graph

We now remove the restriction of preselected features and derive directly from the dynamics which distinctions must be retained to avoid confusing future fields. The analyzed finite reachable set is the same as in Sections 3–5 and contains:

- 56,536 transitions;
- 2,562 distinct current fields; and
- 11,202 distinct $(a, \Delta W)$ states.

Consider a field-only map

$$B : \Delta W \longrightarrow \mathcal{C}$$

and the observation

$$P_B(X_t) = (a_t, B(\Delta W_t)).$$

The requirement is that

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

be single-valued over all 56,536 observed transitions. In other words, the same action and the same field class must never produce different next fields.

Consider two current fields ΔW_i and ΔW_j . If both are observed under a common action a and produce different next fields $\Delta W'_i \neq \Delta W'_j$, they cannot be assigned to the same field class. If

$$B(\Delta W_i) = B(\Delta W_j),$$

then the identical observation

$$(a, B(\Delta W_i)) = (a, B(\Delta W_j))$$

would lead to two different next fields, violating single-valued exact prediction.

We therefore construct an incompatibility graph G with the 2,562 distinct current fields as vertices. An edge joins two fields if they produce different next fields under at least one common observed action. A field-only partition preserving exact prediction is then equivalent to a proper coloring of G , because adjacent vertices must receive different class labels.

It follows that the minimum number of field classes is

$$|\text{Im } B_{\min}| = \chi(G),$$

where $\chi(G)$ is the chromatic number of the incompatibility graph. Graph coloring is computationally difficult in general, so a color count returned by a heuristic does not by itself establish this minimum. In the present finite instance, exactness is obtained by two independent constructions: a proper coloring supplies an upper bound, while an anchor-action decomposition reduces the lower-bound problem to induced subgraphs whose chromatic numbers can be certified exactly. The matching bounds, rather than an assumption about heuristic optimality, are what determine $\chi(G)$.

This formulation does not introduce a new general state-minimization problem. It is connected to compatibility-based minimization of partially specified input-output behavior, particularly a depth-1 partial Mealy-machine formulation of incompletely specified finite-state machine minimization. The objective here is to obtain the concrete exact solution produced when this established problem form is applied to the finite reachable dynamics of ABCW.

Constructive upper bound

We first color the incompatibility graph using DSATUR (Br  laz 1979). DSATUR is used here to construct an upper bound, not to establish optimality. It produces a valid proper coloring with 692 colors and hence establishes

$$\chi(G) \leq 692.$$

The number returned by a coloring algorithm is not, by itself, sufficient to verify the original prediction requirement. We therefore interpret the resulting coloring as the field-only map

$$B_{692}(\Delta W)$$

and map it back to all 56,536 original transitions. We then directly test whether

$$(a, B_{692}(\Delta W)) \longrightarrow \Delta W_{t+1}$$

is single-valued. The result is:

- field classes: 692;
- distinct (a, B_{692}) states: 8,084;
- nondeterministic observed states: 0; and
- one-step complete-field prediction: 100%.

Thus, a field-only partition exists that compresses

$$2562 \longrightarrow 692$$

while preserving exact one-step prediction of the complete field. The class-count-based compression rate is

$$C_{\Delta W} = 1 - \frac{692}{2562} \simeq 0.7299,$$

or approximately 72.99%. The 11,202 complete $(a, \Delta W)$ observations are reduced to 8,084 distinct $(a, B_{692}(\Delta W))$ observations.

Therefore, 692 is an upper bound:

$$\boxed{\chi(G) \leq 692}.$$

At this stage, however, a proper coloring with 691 or fewer colors has not been ruled out. Establishing exact minimality requires an independent lower bound of at least 692.

Anchor lower bound

To construct a lower bound, we fix one action condition. We choose the anchor action

$$a = (-, -, -, -, -).$$

Under this action, 1,239 current fields are observed, and they produce 625 distinct next fields. The lower-bound argument does not assume that these anchor groups cover all 2,562 fields. It uses only the induced subgraph on the 1,239 vertices actually observed under the anchor action.

Two current fields producing different next fields under the same anchor action cannot belong to the same field class. Partition the 1,239 fields into 625 anchor future groups

$$V_1, \dots, V_{625}$$

according to their next field under the anchor action. Every pair of vertices belonging to different groups is adjacent in G .

Lemma 1 — Additivity across anchor groups

For two distinct anchor future groups V_i and V_j , with $i \neq j$, let

$$u \in V_i, \quad v \in V_j.$$

By construction, u and v produce different next fields under the anchor action. Therefore,

$$\{u, v\} \in E(G),$$

and every two distinct anchor future groups form a complete join. Consequently, different anchor groups cannot share a color in any proper coloring.

Let

$$G_i = G[V_i]$$

be the subgraph induced by group V_i . The chromatic number of the anchor-observed induced subgraph is then additive:

$$\chi\left(G\left[\bigcup_{i=1}^{625} V_i\right]\right) = \sum_{i=1}^{625} \chi(G_i).$$

Because every group requires at least one color,

$$\chi(G) \geq 625.$$

Combining this with the constructive upper bound gives

$$\boxed{625 \leq \chi(G) \leq 692}.$$

The value 625 captures only the distinctions exposed by the anchor action. Two fields that lead to the same anchor future may still lead to different futures under another action condition. Such fields must be separated when all observed actions are considered simultaneously.

We therefore inspect the incompatibilities remaining within each of the 625 anchor future groups. The result is:

- 575 groups with no internal conflicts; and
- 50 groups with internal conflicts.

For the 575 conflict-free groups, $\chi(G_i) = 1$. The chromatic numbers of the remaining 50 induced subgraphs must be determined exactly.

A global sign-flip symmetry provides a useful redundancy and consistency check for the internal-conflict data but is not used in the lower-bound proof; it is documented separately in [Appendix A](#).

Structural origin and exact refinement of the 50 difficult anchor groups

The 50 nontrivial anchor groups allow the lower-bound refinement to be connected directly to the field-update rule. We first characterize how two distinct current fields that are indistinguishable under the anchor action can become incompatible under another observed action.

Because $n = 5$, every non-unanimous binary action profile has a strict minority and majority. Hence $u_i(a) \in \{-1, +1\}$, with $+1$ for a winner and -1 for a loser; under the unanimous anchor action $a^* = (-, -, -, -, -)$, the tie convention makes every agent a loser. All experiments use $\eta = 1$. For an existing off-diagonal edge,

$$w'_{ij} = \max(0, w_{ij} + u_i(a)).$$

Since $w^*_{ij} = 1$ for $i \neq j$, this is equivalently

$$\Delta w'_{ij} = \max(0, \Delta w_{ij} + 1 + u_i(a)) - 1. \quad (18)$$

Lemma 2 — Anchor clipping ambiguity

Let ΔW and $\widetilde{\Delta W}$ be two current fields in the same anchor future group V_k . Then, for every off-diagonal coordinate $i \neq j$,

$$\Delta w_{ij} \neq \widetilde{\Delta w}_{ij} \implies \{\Delta w_{ij}, \widetilde{\Delta w}_{ij}\} = \{-1, 0\}. \quad (19)$$

Proof. Under a^* , every agent has payoff -1 , so Eq. (18) reduces to

$$\Delta w'_{ij} = \max(0, \Delta w_{ij}) - 1.$$

Two fields in the same anchor future group have the same next field under a^* , hence

$$\max(0, \Delta w_{ij}) = \max(0, \widetilde{\Delta w}_{ij}).$$

In the enumerated system the initial weights are integer-valued and $\eta = 1$, so every reachable w_{ij} , and hence every reachable Δw_{ij} , is integer-valued. Together with $w_{ij} \geq 0$ and $w_{ij}^* = 1$, this gives $\Delta w_{ij} \in \{-1, 0, 1, 2, \dots\}$ for every off-diagonal coordinate. On this discrete domain the map $x \mapsto \max(0, x)$ identifies exactly the two distinct admissible values -1 and 0 and is injective otherwise. The stated implication follows. $\square$

In absolute-weight coordinates, the same ambiguity is the clipping of $w_{ij} = 0$ and $w_{ij} = 1$ to the same value under a losing update.

For an action profile a , define the winning-agent set

$$R(a) = \{i \in V : u_i(a) = +1\}.$$

Proposition 1 — Winner-row revelation criterion

Let $\Delta W, \widetilde{\Delta W} \in V_k$ belong to the same anchor future group, and let a be an action condition observed with both current fields in the finite dataset. Denote their next fields under that action by $\Delta W'$ and $\widetilde{\Delta W}'$. Then

$$\Delta W' \neq \widetilde{\Delta W}' \iff \exists i \in R(a), \exists j \neq i : \Delta w_{ij} \neq \widetilde{\Delta w}_{ij}. \quad (20)$$

Proof. Suppose first that the right-hand side holds. For the corresponding winning agent i , $u_i(a) = +1$, and Eq. (18) gives

$$\Delta w'_{ij} = \Delta w_{ij} + 1, \quad \widetilde{\Delta w}'_{ij} = \widetilde{\Delta w}_{ij} + 1,$$

because $\Delta w_{ij} \geq -1$. A difference in that current-field coordinate therefore remains a difference in the next field.

Conversely, suppose $\Delta W' \neq \widetilde{\Delta W}'$. Then some off-diagonal coordinate $i \neq j$ differs. If $i \notin R(a)$, then $u_i(a) = -1$, and both coordinates are updated by $x \mapsto \max(0, x) - 1$. Because the same-anchor-group equality $\max(0, \Delta w_{ij}) = \max(0, \widetilde{\Delta w}_{ij})$ holds for every row, and the losing-row update is this same map for every action under which agent i loses, losing rows agree under every such action, not only under a^* . Hence the two next-field coordinates must agree, a contradiction. Thus $i \in R(a)$. On a winning row the update is translation by $+1$, so differing next-field coordinates also imply differing current-field coordinates. $\square$

Corollary 1 — Within-anchor edge criterion

For a current field ΔW , let $\mathcal{A}(\Delta W)$ be the set of action conditions observed with it in the finite dataset. Two fields $\Delta W, \widetilde{\Delta W} \in V_k$ are adjacent in $G_k = G[V_k]$ if and only if there exist

$$a \in \mathcal{A}(\Delta W) \cap \mathcal{A}(\widetilde{\Delta W}), \quad i \in R(a), \quad j \neq i,$$

such that

$$\Delta w_{ij} \neq \widetilde{\Delta w}_{ij}. \quad (21)$$

Thus the anchor action can hide a current-field distinction only through the clipping ambiguity of Lemma 2, while a co-observed action exposes that distinction precisely when a row containing it belongs to a winning agent. For clarity, the reported 10,094 checks count unordered pairs of distinct fields within each anchor group once for every action observed with both fields. Equivalently, the number of comparisons is

$$\sum_k \sum_{\{W, \widetilde{W}\} \subset V_k} |\mathcal{A}(W) \cap \mathcal{A}(\widetilde{W})| = 10,094, \quad (22)$$

where pairs with no co-observed action contribute zero. Exhaustive evaluation of these comparisons in the stored dataset agreed with the analytic criterion in every case; this enumeration is an implementation check, not the proof of the proposition.

Proposition 2 — Exact clique–chromatic structure

For the finite reachable dataset analyzed here, exactly 575 of the 625 induced anchor graphs are edgeless and therefore have $\chi(G_i) = 1$. The remaining 50 contain internal incompatibilities. Exact finite computation gives

$$\chi(G_i) = \omega(G_i) \quad (22)$$

for every one of these 50 induced graphs, where $\omega(G_i)$ is the clique number. Their common clique/chromatic numbers are distributed as

k	2	3	4	7
$\#\{i : \chi(G_i) = \omega(G_i) = k\}$	39	8	2	1.

(23)

For each difficult group, an explicit clique supplies the lower bound $\omega(G_i) \leq \chi(G_i)$, while an explicit proper coloring supplies the matching upper bound $\chi(G_i) \leq \omega(G_i)$. Equality is therefore certified group by group rather than inferred from a greedy coloring heuristic. The unique $k = 7$ group has 44 vertices and 224 internal conflict edges and contains a K_7 , so seven pairwise incompatible reachable fields already force seven predictive classes within that anchor group.

Remark on computational status. Proposition 2 is an instance-specific exact computational result. For each difficult group, the lower and upper bounds are supplied by matching finite certificates—an explicit clique and an explicit proper coloring—rather than by treating the output of a heuristic as proof of optimality. Maximum cliques are obtained by an exact Bron–Kerbosch-style search (Bron and Kerbosch 1973).

Equation (22) is an exact property of these 50 induced graphs in the present finite dataset. We do not claim that $\chi = \omega$ holds for arbitrary ABCW anchor graphs, nor does equality for these graphs alone establish that they are perfect graphs.

Relative to one color for each anchor future group, the exact additional number of colors is therefore

$$\begin{aligned} \sum_{i=1}^{625} (\chi(G_i) - 1) &= 39(2 - 1) + 8(3 - 1) + 2(4 - 1) + (7 - 1) \\ &= 39 + 16 + 6 + 6 \\ &= 67. \end{aligned} \quad (24)$$

By Lemma 1, colors cannot be shared across distinct anchor future groups, so the anchor-observed induced subgraph requires

$$\sum_{i=1}^{625} \chi(G_i) = 625 + 67 = 692. \quad (25)$$

The direct cross-check is

$$575 \cdot 1 + 39 \cdot 2 + 8 \cdot 3 + 2 \cdot 4 + 1 \cdot 7 = 692.$$

Hence the complete incompatibility graph satisfies

$$\boxed{\chi(G) \geq 692}.$$

The lower bound now has a dynamical interpretation in addition to its computational certificate. Anchor clipping hides only $-1/0$ field-deviation distinctions; alternative co-observed actions reveal such distinctions through winning rows; the resulting incompatibility edges form, in each difficult group, cliques large enough to attain the exact chromatic number. Thus the additional 67 classes are forced by explicit sets of pairwise incompatible reachable fields rather than by an excess of a coloring heuristic.

Figure 3 summarizes the matching upper and lower bounds and the $625 + 67$ decomposition. Table 4 records the exact contribution of each anchor-group chromatic number.

Table 4. Exact lower-bound decomposition

Required colors per anchor group	Number of anchor groups	Contribution to total classes	Additional colors beyond one per group
1	575	575	0
2	39	78	39
3	8	24	16
4	2	8	6
7	1	7	6
Total	625	692	67

Main result

Section 6.2 constructs a valid 692-coloring of the incompatibility graph on all 2,562 fields, establishing

$$\chi(G) \leq 692.$$

Direct evaluation of this coloring on all 56,536 original transitions verifies that

$$(a, B_{692}(\Delta W)) \longrightarrow \Delta W_{t+1}$$

is single-valued throughout the observed dataset.

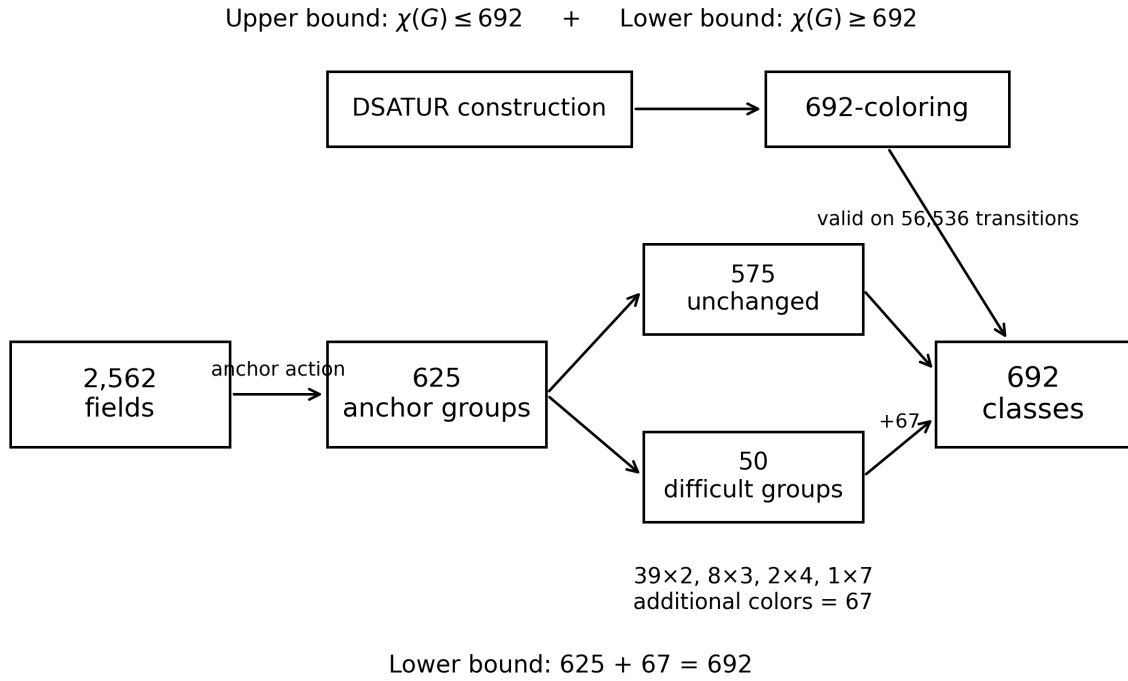


Figure 3: Exact minimum predictive partition. The constructive 692-coloring provides the upper bound. The anchor action yields 625 groups; exact coloring of the 50 internally conflicting groups requires 67 additional colors, producing a matching lower bound of 692.

The anchor lower-bound analysis decomposes the 1,239 fields observed under the anchor action into 625 future groups. Complete joins across different groups, the analytic winner-row criterion within groups, and the exact clique-chromatic certificates for the 50 difficult groups establish

$$\chi(G) \geq 692.$$

The bounds coincide:

$$\boxed{\chi(G) = 692}.$$

Main Result.

For the 56,536 observed transitions and 2,562 distinct current fields of the five-agent ABCW system, consider a field-only map

$$B : \Delta W \longrightarrow \mathcal{C}$$

such that

$$(a_t, B(\Delta W_t)) \longrightarrow \Delta W_{t+1}$$

is single-valued over all observed transitions. The exact minimum number of field classes is

$$\boxed{|\text{Im } B_{\min}| = 692}.$$

It is therefore unnecessary to retain all 2,562 distinct current fields for the specified task. Exact prediction of the complete one-step-ahead field permits the reduction

$$2562 \longrightarrow 692,$$

with the class-count-based compression rate

$$1 - \frac{692}{2562} \simeq 72.99\%.$$

Any partition into 691 or fewer classes fails even on the anchor-induced subgraph used in the lower-bound argument. It must assign the same class to at least one pair of fields that produce different next fields under a common action condition, thereby violating exact prediction.

Thus, 692 is not merely the size of a high-performing candidate partition. It is the minimum number of classes in a field-only partition that preserves exact one-step, action-conditioned complete-field prediction on the observed finite reachable set.

The result also clarifies the contrast with the natural-feature baselines. Among the 511 tested feature combinations, none achieved both genuine field compression and exact prediction. This was not because exact prediction intrinsically required nearly complete microscopic field identification. Once the restriction to preselected features is removed and the partition is derived from future incompatibilities, the 2,562 fields can be reduced to 692 classes. Hence,

$$\boxed{\text{human-designed features} \neq \text{minimum distinctions required for prediction}}.$$

The exact claim is subject to the Scope statement in the Introduction; changing the reachable set, retained variables, target, or prediction horizon defines a different minimization problem and may change the minimum.

The theorem determines the chromatic number, and hence the minimum class count of 692. The concrete map B_{692} verified above is one representative minimum partition obtained from the proper coloring constructed by DSATUR. A different 692-coloring—including one constructed from the anchor decomposition—may assign fields to classes differently. We therefore claim uniqueness only of the minimum class count for this incompatibility graph, not uniqueness of the minimizing map or partition.

The search for the minimum number of classes is now complete. Determining that the reduction is $2562 \rightarrow 692$, however, is not the same as understanding what the 692 classes represent. [Section 7](#) therefore examines which fields are merged within the classes, which distinctions are exposed by action conditions, how the additional 67 distinctions are localized, and what symmetries are present in the partition structure.

Structure and Interpretation of the Minimum Partition

The preceding analysis established that 692 is the minimum number of classes required to predict the next field $\Delta W'$ uniquely from the current field ΔW conditioned on the action a , over the 2,562 observed fields.

We now examine the empirical structure of the 692 classes rather than restate the minimality proof. The objective is not to assign an immediately interpretable meaning to every class, but to show how the certified minimum is realized by genuine merging and by localized refinement of the anchor groups.

Class-size structure and genuine merging

The original observation set contains 2,562 distinct values of ΔW . The identity observation

$$P_{\text{id}}(\Delta W) = \Delta W$$

distinguishes all of them and therefore has 2,562 classes. By contrast, the minimum predictive partition reduces the distinction count to

$$2562 \longrightarrow 692.$$

In terms of field-class count, approximately 73% of the original distinctions can be discarded without losing one-step field prediction on the observed dataset.

This compression is not produced solely by a small number of exceptionally large classes. Of the 692 classes:

- 352 contain a single ΔW ;
- 340 contain multiple distinct values of ΔW ; and
- the largest class contains 65 distinct values of ΔW .

Many fields must therefore remain individually distinguished, but genuine merging is also widespread. Microscopically different fields can often be treated as identical for the specified prediction task. The result provides a concrete finite example in which

microscopically distinct

does not imply

must remain distinct for prediction.

Anchor localization of the refinement

Section 6.4 establishes the exact $625 + 67 = 692$ lower-bound decomposition and identifies the local dynamical mechanism generating within-anchor conflicts. Here we retain only its structural implication for interpreting the final partition. Of the 625 anchor future groups, 575 remain unsplit when all observed action conditions are considered, while all additional refinement is localized in the other 50. Those 50 groups produce 117 predictive subclasses in total, so their net contribution beyond one class per anchor group is

$$117 - 50 = 67.$$

Thus the minimum partition has the qualitative form

large base partition + localized action-conditioned refinement.

This localization is not apparent from a lookup table of 692 labels. Figure 4 summarizes it together with the final composition of 352 singleton and 340 multi-field classes.

Geometric proximity versus predictive equivalence

We next ask whether fields assigned to the same class are also similar in an ordinary geometric sense. If the minimum predictive partition resembled a simple geometric clustering, nearby values of ΔW would be expected to share classes, while distant fields would tend to be separated.

Inspection of the partition does not support this expectation in general. Fields that differ substantially in their microscopic representations can belong to the same predictive class, while nearby fields can be assigned to different classes.

The partition therefore does not preserve the geometry of the ΔW space itself. It preserves the action-conditioned future-response structure

$$\Delta W_t \mapsto (a \mapsto \Delta W_{t+1}).$$

In this predictive sense, two fields are “close” not because their matrix entries differ only slightly, but because they produce indistinguishable future responses under the relevant observed action conditions.

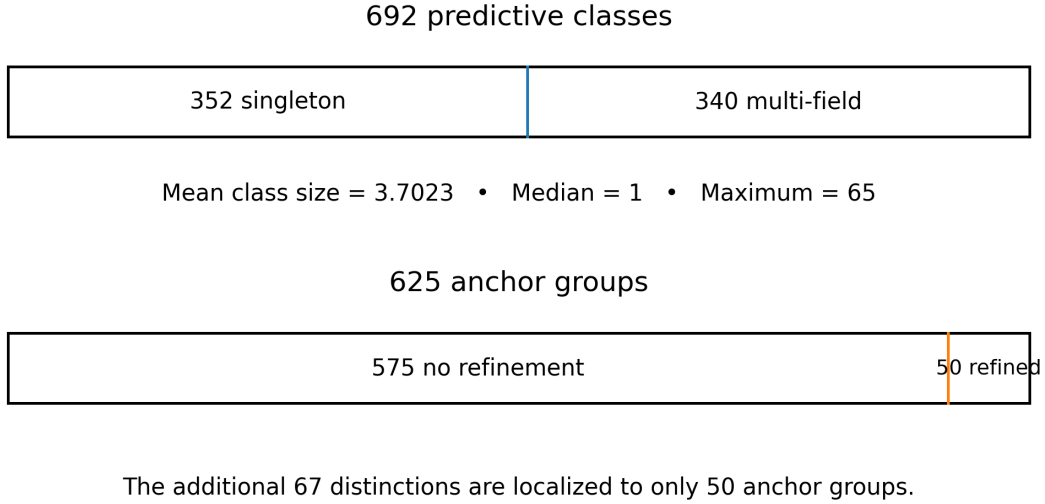


Figure 4: Internal structure of the 692-class partition. Top: composition of the 692 predictive classes (352 singleton and 340 multi-field classes), with verified mean, median, and maximum class size. Bottom: localization of all additional refinements to 50 of the 625 anchor groups.

Why the partition is not yet an interpretable macrovariable

The existence of a 692-class minimum partition does not establish that a natural macrovariable of ABCW has been discovered.

First, the partition is constructed from an observed finite reachable set. Second, its target is restricted to the one-step-ahead field ΔW_{t+1} . Third, the classes are defined by future responses, and it has not been shown that they can be expressed through simple physical or network quantities. Fourth, exact determination of the minimum class count does not imply that the class labels or the partition realizing that count are unique: the DSATUR-derived B_{692} is a verified representative optimum, not a canonical quotient.

The result directly supports only the following claim:

In the observed finite ABCW system, not all 2,562 microscopic field distinctions are required to preserve exact one-step, action-conditioned field responses; 692 predictive distinctions are necessary and sufficient.

An intermediate descriptive level

The 692 classes occupy an intermediate descriptive level between a complete microscopic representation and a small set of simple macroscopic statistics. They retain the distinctions required for a specified future while discarding distinctions that are irrelevant to that target on the observed dataset.

Whether this intermediate description persists under altered systems, reachable sets, or prediction targets is left open; the formal scope and limitations are collected in [Section 8, Scope and limitations](#).

Discussion

The exact minimum and its class structure were established in [Section 6](#) and [Section 7](#). This section therefore focuses on implications, connections to neighboring frameworks, the dynamical origin of the additional refinement, and the limits of the claim.

Hand-designed observables and dynamics-derived partitions

Section 5 showed a sharp contrast: simple network summaries can be highly predictive, yet none of the 511 tested feature combinations achieved both nontrivial field compression and exact prediction. Section 6, by deriving distinctions from future-response conflicts instead of selecting features in advance, reduces 2,562 fields to 692 exact predictive classes.

The point is not that natural features are generally unsuitable. Rather, the exact partition supplies a benchmark against which interpretable coarse-grainings can be compared. On this fully enumerable finite set, predictive sufficiency and the minimum class count are independently certified; a future macrostate-discovery method can therefore be judged against known ground truth rather than only against a heuristic score.

Relation to state aggregation and predictive-state ideas

The detailed comparisons with lumpability, bisimulation, causal states, PSRs, and incompletely specified finite-state machines are given in Section 2. Their common relevance here is that aggregation is defined by what future distinctions must survive. The present construction occupies a deliberately narrow point in that landscape: it preserves a specified external one-step target on partially specified finite input-output data, while retaining the action and not requiring the reduced field label to form recursively closed dynamics. That restriction is what turns the minimum into an exact graph-coloring problem.

The 692-class partition should therefore be read neither as a new general state-minimization formalism nor as a causal-state or PSR construction. Its value is as an exactly solved finite instance in which the prediction task, admissible mergers, and minimum number of classes can all be inspected explicitly.

The structure behind 692

The lower-bound decomposition now has both a local dynamical explanation and an exact global graph-theoretic realization. Under the unanimous anchor action, the losing update clips the two admissible field-deviation values -1 and 0 to the same next value. Lemma 2 shows that this is the only coordinate-level ambiguity possible within an anchor future group. Proposition 1 then shows that another co-observed action exposes such a hidden distinction precisely when the differing coordinate lies in a winning agent’s row. Corollary 1 therefore characterizes the within-anchor incompatibility edges directly from the update rule.

This mechanism is localized in the reachable dataset. Of the 625 anchor future groups, 575 have no internal incompatibility edge. The remaining 50 contain the distinctions that become prediction-relevant under alternative action conditions. Proposition 2 shows, by matching clique and coloring certificates, that each of these 50 induced graphs satisfies $\chi(G_i) = \omega(G_i)$ in the present instance, with the 39/8/2/1 distribution over chromatic numbers 2, 3, 4, and 7. Thus the additional 67 classes are forced by explicit sets of pairwise incompatible reachable fields:

$$39(2 - 1) + 8(3 - 1) + 2(4 - 1) + (7 - 1) = 67.$$

The structural chain is therefore

$$\begin{aligned} \text{clipping} &\longrightarrow \text{hidden reachable distinctions} \longrightarrow \text{winner-row revelation} \\ &\longrightarrow \text{incompatibility edges} \longrightarrow \text{clique-certified refinement.} \end{aligned}$$

This does not explain why the reachable set produces exactly the 39/8/2/1 distribution, nor does it imply $\chi = \omega$ for arbitrary ABCW anchor graphs. It does, however, explain why the refinement beyond the 625 anchor classes is required by the dynamics rather than being an artifact of a coloring procedure. The remaining unexplained structure is therefore narrower: it concerns the global organization of these locally generated conflicts in the particular reachable set.

Scope and limitations

The exact value 692 belongs to the five-agent model and to the finite reachable set generated from the four initial fields used here; it cannot be extrapolated directly to other agent numbers, initial conditions, or ABCW systems. The current action is retained and only ΔW_t is compressed, so the result is not a minimum joint representation of action and field.

The target is the complete one-step-ahead field ΔW_{t+1} . The 692 classes are not shown to support multistep prediction or closed recursive reduced dynamics. Moreover, the input-output relation is only partially specified: not every action is observed for every field, so the minimum applies to the observed reachable set and makes no claim about unobserved combinations.

Finally, exact prediction is imposed as a hard constraint. Simpler representations may be scientifically preferable when small prediction errors are acceptable. Nor does the minimum class count establish a unique minimizing partition or a concise analytic/macrosopic representation of its classes. These are distinct questions from the minimum-count problem solved here.

Future directions

Natural extensions are to increase the prediction horizon, jointly compress action and field, relax exactness to study the prediction–compression–representation-cost trade-off, and vary the agent number, initial fields, and reachable set. For multistep targets, compatibility must become recursive because merged records must also have compatible successors; this reconnects the construction with the closure conditions of incompletely specified finite-state-machine minimization.

Two focused questions arise directly from the present result: whether the 692 classes admit a concise interpretable representation, and what reachable-set structure produces the observed 39/8/2/1 distribution of nontrivial anchor chromatic numbers. The instance-specific equality $\chi = \omega$ in all 50 difficult anchor graphs motivates, but does not establish, any broader perfect-graph or $\chi = \omega$ claim.

Conclusion

For the specified finite five-agent ABCW prediction problem, the minimum field-only partition preserving exact prediction of the complete one-step-ahead field has

$$|\text{Im } B_{\min}| = \chi(G) = 692.$$

A constructive 692-coloring gives the upper bound, while the independent anchor decomposition gives the matching lower bound $625 + 67 = 692$. The additional 67 classes have a direct dynamical origin: the unanimous losing update hides reachable $-1/0$ distinctions by clipping, and alternative co-observed actions reveal them precisely through winning rows. Matching clique and coloring certificates establish the exact refinement in each of the 50 nontrivial anchor groups.

The comparison with 511 natural-feature combinations supplies a complementary baseline. Highly predictive summaries need not coincide with the minimum distinctions required by a specified future, while a dynamics-derived exact partition need not immediately be interpretable as a simple macrovariable. Within the limitations stated in Section 8, the result provides a finite ground-truth example of the basic principle that being microscopically distinct is not the same as needing to remain distinct for prediction.

Data and Code Availability

The complete transition dataset, source code, exact-coloring certificate, lower-bound reconstruction, natural-feature analysis, anchor-structure analysis, maximum-clique witnesses and matching coloring certificates for the 50 difficult groups, the exhaustive winner-row criterion cross-check, and automated verification scripts are available at <https://github.com/youque8/abcw-paper-reproducibility>. The repository release corresponding to this manuscript is version 1.1.0. The core numerical chain can be checked by running `python scripts/verify_all.py`; the full recomputation, including all 511 natural-feature combinations and exact re-solving of the anchor refinements, is available through `python scripts/verify_full.py`.

Global action sign-flip symmetry

Under the global action flip $a \mapsto -a$, minority-game payoffs and therefore the one-step field update are unchanged. Hence a and $-a$ witness the same field-conflict edges, and the 30 non-anchor actions form 15 global-flip orbits. This symmetry is not used in the 692 lower bound; it is only a redundancy reduction and independent consistency check.

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